



21 (a) $(3 - \sqrt{5})(2 + 3\sqrt{5}) = a + b\sqrt{5}$

Find the value of a and the value of b .

$$3(2 + 3\sqrt{5}) - \sqrt{5}(2 + 3\sqrt{5})$$

$$6 + 9\sqrt{5} - 2\sqrt{5} - 15$$

$$-9 + 7\sqrt{5}$$

$$a = \dots -9 \dots$$

$$b = \dots 7 \dots$$

[2]

- (b) Rationalise the denominator.
Write your answer in its simplest form.

$$\frac{6}{\sqrt{2}}$$

$$\frac{6 \times \sqrt{2}}{\sqrt{2} \times \sqrt{2}}$$

$$\frac{3 \cancel{\sqrt{2}}}{\cancel{\sqrt{2}}} = 3\sqrt{2}$$

$$\dots 3\sqrt{2} \dots [2]$$

22 Solve.

$$\frac{2}{x-1} = \frac{x}{x+2}$$

$$2(x+2) = x(x-1)$$

$$(x-4)(x+1) = 0$$

$$2x + 4 = x^2 - x$$

$$\underline{x = 4} \text{ OR } \underline{x = -1}$$

$$0 = x^2 - x - 2x - 4$$

$$0 = x^2 - 3x - 4$$

$$x^2 - 4x + x - 4$$

$$x(x-4) + 1(x-4)$$

$$x = \dots 4 \dots \text{ or } x = \dots -1 \dots [5]$$